Question and design

Does model width improve contact accuracy at the same one-epoch exposure? Baseline: exp277, native plus MPNN corpus, 24 layers. Hidden width 2048 -> 4096; MLP width 8192 -> 16384. Keep 32 query / 8 KV heads; head width 64 -> 128. Keep data, optimizer schedule, seeds and global batch fixed.

Architecture and smoke verified

5,861,984,256 parameters, versus exp277's 1,471,374,336 (3.984x). Architecture, tokenizer, data and recovery checks passed. 64-H100 smoke: ten updates, train loss 8.56948, eval loss 8.03575. 2.821 seconds/update; native and HF checkpoints saved with tokenizer. Compressed smoke schedule was spiky; this is an execution check.

Current status — October 23

Credentials refreshed; fsspec and TensorStore reads verified. Original a04 submitted at 10:34 EDT; restored checkpoint 75,031 -> step 75,032. Fork a04 submitted at 10:41 EDT; source checkpoint 74,178 -> step 74,179. Both use four nodes / 32 H100s, batch, with independent identities and paths. Both are training: original first resumed update 10:43 EDT; fork 10:51 EDT. Fork step 74,190: LR 0.00099847547, approximately 5.59 seconds/update. Fixed credential forwarding and remote optimizer class serialization; 14 tests pass. Latest pre-restart full validation: original 3.09826 at step 73,990.

TPU trial succeeded

Interactive v5p-64: 32 chips across 8 VMs; all 20 updates completed. Finished October 16; no worker failures or preemptions. Native and HF step-19 checkpoints verified; tokenizer included. Median steps 5–19: 7.40 s/update, about 32% longer than GPU steps. Short native-only smoke; throughput is preliminary, not cost-normalized. Production continues on GPUs. No bulk transfer or migration performed.

Matched validation: no width benefit yet

Same native + MPNN corpus, global batch, validation and optimizer schedule. Wider model led early; default overtook it between steps 33,824 and 35,938. Default has lower loss at all 18 subsequent full evaluations. At step 71,876: default 3.08621; wider 3.09976. Gap +0.01354 nats/token, or 1.36% higher perplexity. Gap narrowed from its +0.01685 peak at step 61,306. Single seed, incomplete epoch; baseline LR retained without retuning. Both losses keep falling. No contact-accuracy evaluation for the wider model yet.

Other architectures: context, not a common leaderboard

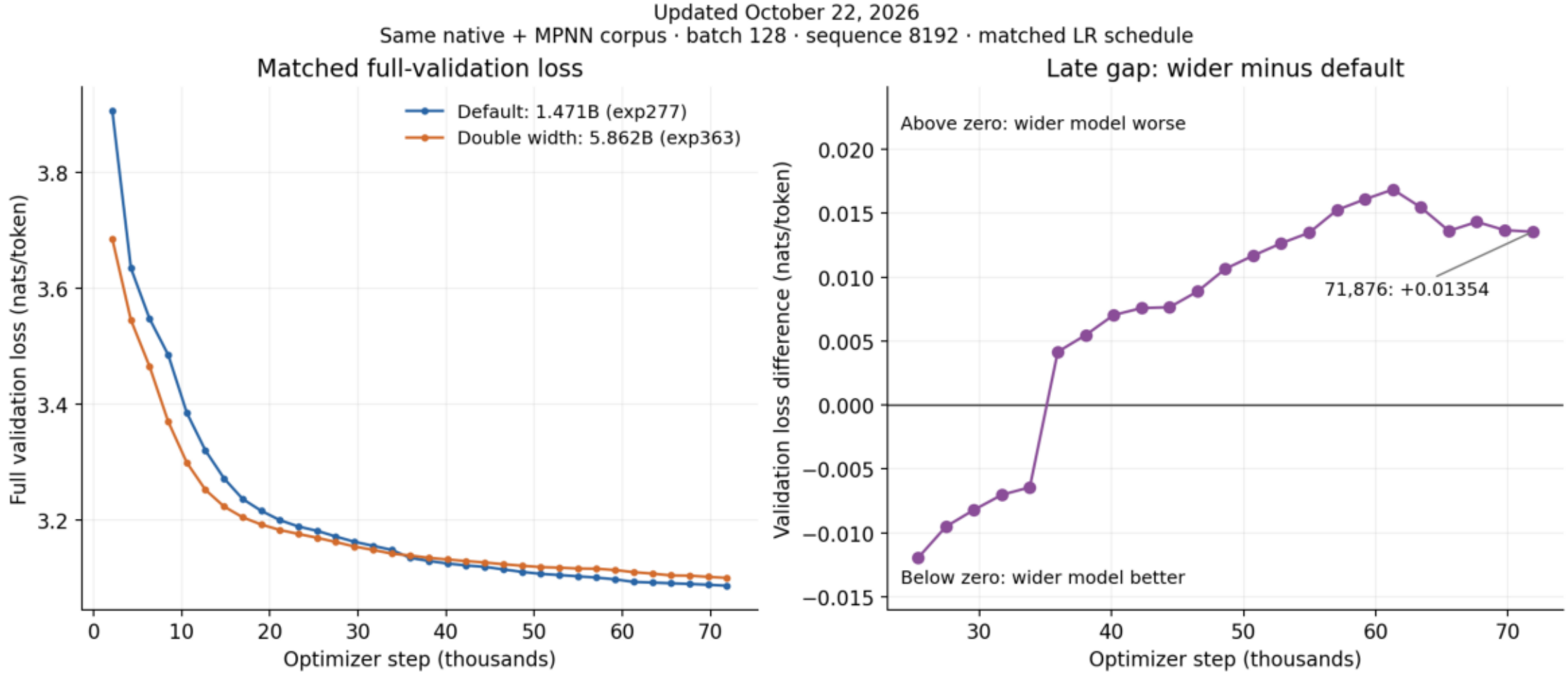
exp108 depth-scaled 2.93B: all three LR trials unstable around 6.8k updates. exp108 width-scaled 2.78B: all three trials finished; older recipe and validation. exp262 NoPE + smear: final loss 3.00207 versus its control's 2.97454. Its early advantage reversed, illustrating why interim rankings can mislead. exp157 learned RoPE deltas: corrected CE 3.0654 versus vanilla 3.0815. Only exp277 provides the same native + MPNN recipe as the current wider run. Raw historical validation losses cannot be pooled across cache/logging paths.

Parallel LR fork — October 22

Original schedule retains its 0.001 plateau and final cooldown when resumed. Fork restores checkpoint 74,178, resuming at absolute step 74,179. Weights, Adam moments/counters, RNG, data position and augmentation are preserved. Linear reduction to 0.000707 over 2,114 updates; target reached at step 76,293. Hold until original final cooldown at step 213,076; final LR target 0.0000707. Same 266,345-update horizon, separate W&B and checkpoint identities. Source copy and counters verified; 14 local configuration/restore/serialization tests passed. October 22 a02 failed before training; October 23 credentials refreshed. Retry a03 exposed CLI-class serialization; moved optimizer into an importable module. Corrected a04 is training; first nine LRs match the schedule within 1.14e-10. First loss at 74,179 exactly matches the original; no new full validation yet.

training_comparison.png

Full-validation observations through step 71,876, exported October 22, 2026. Exact-step comparison; no interpolation or smoothing. Same logical training recipe, different hardware/gradient accumulation; single seed, incomplete epoch. The right panel zooms into observations after step 25,000.




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$ plot_training_comparison.py
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